Room plan: stars and wheels (hop around a ring)

Tentative: the next circle after bridges on November 18, if the circle meets again before the end of fall quarter (proposed by Claude, November 3, 2026; moved back a week November 5 when November 11 turned out to be a holiday; not yet confirmed by James). Week 4, Stars and wheels: packets for K–1, 2–3 and 4–5, and the facilitator guide, whose one-hop launch this plan uses, acted out on chairs. One theme at three tables by band, one adult hosting each table all hour. Unpiloted, at the circle and at home.

Why this follows bridges

Bridges asked whether one walk could cross every bridge, and in some towns a counter was left behind however you walked. Today the walk has no choices at all: every move is the same hop around a ring of dots. Sometimes one start reaches every dot and draws a star; sometimes the pencil is back early and has to start again at a bare dot. It is the same kind of question (can one trip reach everything, and if not, how many trips?), and again the answer is a number the children can learn to find before they draw. It also closes a loop with November 4's shuffle machines: a hop is a machine that moves every cube the same number of places, and its loops are today's starts, all the same length.

For adults: number the n dots 0 to n − 1. Hopping k from any dot comes back after n ÷ g hops, where g is the biggest number that goes into both n and k (gcd(n, k)). So it takes exactly g starts, each a copy of the first turned one dot along, and one start reaches every dot exactly when g = 1. On a prime number of dots every hop does; on any other number some hop fails. Hop k and hop n − k draw the same lines the other way round (on 12 dots, hop 8 is hop 4 backwards). Why, in the children's terms (guide, 4–5 Problem 7): the first start lands only on the g-times table, so the dots in between each need their own start; and every start is the first one turned, so all starts have the same number of dots and share the n dots out in equal piles. The code wheel is the same ring with 26 places: a setting is a hop, two settings in a row add (going round past Z), a setting and its undo add to 26, and a setting takes A through all 26 letters exactly when it moves letters an odd number of places and is not N (13 places). K–1's picture ring is a ring of 6, where a hop and the hop that undoes it add to 6. What the children should find: K–1, which hops land on every dot on rings of 4 to 7, and starting again at a bare dot; 2–3, a guess for the number of starts with a reason ("hop 3 on 12 only lands on 0, 3, 6 and 9"), and that the popular guess "starts = hop" fails for hop 5 on 12; 4–5, the rule, tested on big numbers, a reason, and that two settings make one. The opener and the finale are this ring made of people: seats counted clockwise are dots, and a note passed 2 seats along is a hop of 2.

Prep checklist for Wednesday afternoon

Supplies (James, November 4): room for the 10-foot circle by moving tables; 60 counters in two colours, coloured pencils, whiteboards and markers, and card stock in hand; brass fasteners and yarn ordered. Still to confirm: 8 scissors, 9 six-inch rulers and a pushpin.

Print single-sided on Letter paper at 100% ("Actual size"); "Fit to page" shrinks the wheels and the K–1 dots. The packets are black and white; the opener prints fine either way. The guide's counts assume 4 children at K–1, 4 at 2–3 and 3 at 4–5; the 2–3 counts below lean up to 6, since most of the room is there, and every table has a spare for a child who moves over at the midpoint.

- K–1 packet, 10 pages: 6 copies (4 children, the host, 1 spare), not stapled; the host hands out one page at a time
- Grades 2–3 packet, 9 pages: pages 1–7, 8 copies, stapled (6 children, the host, 1 spare); page 6 again, 6 copies (Problem 7's rings for trying; skip if prep is short); pages 8–9 (the wheels), 8 copies, on card stock if the printer takes it
- Grades 4–5 packet, 11 pages: pages 1–9, 5 copies, stapled (3 children, the host, 1 spare); pages 10–11 (the wheels), 4 copies. The wheels are the same at both tables, so any child can use either set.
- Facilitator guide, 10 pages: one per adult (3 or 4)
- This room plan: one per adult
- Opener sheet, 2 pages, cut in half: page 1 (the main puzzle, two per page) 6 copies, page 2 (four seats on top, six seats below) 2 copies
- Counters, about 61 (guide): two K–1 cups, each with 1 counter of one colour (the hopper) and 7 of

another (markers); 20 loose at each of the other two tables; 1 for the note
- Coloured pencils, six or more colours, 6 sets; pencils; 6-inch rulers, 9; scissors, 8; brass fasteners, 20 (legs bend); a pushpin; small whiteboards and markers, 3, one for the launch board
- November 4's animal signs (BEAR, FOX, OWL, RABBIT, TURTLE), or five new ones in marker, and one sheet with a big star on it for the star seat
- **Added for the finale, not in the guide:** one ball of yarn, any colour (a second colour is nice, not needed). Keep a pair of scissors with it.
- Three table cards and three early-finisher cards, handwritten (text below)
- A timer everyone can see
- The guide's checks (section 2), the day before or that afternoon: the outer wheel measures 5.5 inches across and the inner 4 inches; a counter sits on one K–1 dot without touching the next (pennies if not)
- Pierce the wheel centres: stack the outer-wheel pages and push the pushpin through the centre cross; then the inner-wheel pages
- Spare wheels: 2 (one per older table), cut and joined as in the guide, step 4. The guide asks for 3; make the third only if there is time.

Prep time: about 15 minutes to print and staple, 5 for the guide's checks, 5 to pierce, 8 for two spare wheels, 3 to pack the K–1 cups, 8 to write the cards: about 45 minutes, plus 5 at the room for the chairs and the launch board. If short: skip the second copy of 2–3 page 6, make one spare wheel, and write each table card on one side only (the first-half question) and say the second-half question aloud at the midpoint.

Before the children arrive

- **The ring of chairs**, in open floor with room around it for a sitting circle of about 12 (8 to 10 feet across) for the finale. Five chairs in a circle, facing in. The star sheet on the floor in front of one chair: that is the star seat. The animal signs on the floor beside the chairs. A sixth chair to one side, for the midpoint.
- **The launch board** (guide, step 6), propped by the chairs: a ring of 5 dots as big as the board allows, the top dot filled in, a clockwise arrow inside. The filled dot is the star seat. One counter: the note.
- On every table: opener half-sheets and pencils, so the first child in starts on the puzzle; the table card, first side out; that band's pages face down; coloured pencils and rulers; the counters. The early-finisher card stays with the host. The 2–3 and 4–5 hosts keep scissors, fasteners and wheel pages until the wheel.
- The yarn, scissors and timer by the chairs.
- **Hosts.** One adult per table, all hour (guide: K–1 a parent volunteer, 2–3 the mathematician, 4–5 the organizer). The organizer also leads the launch, the floor moves and the time. A fourth adult floats: the timer, late arrivals, and the 2–3 table while it cuts wheels.
- **Table cards.** Fold a sheet into a tent and write big. The first side faces the children until the midpoint; then the host turns the card round. A child who walks up mid-round reads the card (or the host reads it) and joins a pair.
    - K–1, first side (draw a ring of 5 dots with an arrow): "Hop the counter. Say *one* on the next dot. Does it land on every dot before it is back? If a dot has no line, start again there."
    - K–1, second side (draw the ring of six pictures): "A hop moves every picture along the ring. What does the secret hop do? Which hop changes it back?"
    - 2–3, first side: "Draw hop 3: a straight line to the dot 3 places clockwise. Keep hopping until you are back. A dot with no line? Start again there. How many starts? Can you tell before you draw?"
    - 2–3, second side: "Code wheel. Set to D: inner A next to outer D. CAT goes into code as FDW. To decode, find the code letter on the outer wheel. Crack the codes, then make one for your partner."
    - 4–5, first side: "Each start makes one piece. Find a rule for the number of pieces from the number of dots and the hop, and use it without drawing."
    - 4–5, second side: "Code wheel: crack codes made with settings nobody wrote down. Two settings in a row: which one setting does the same?"
- **Early-finisher cards**, one per table:
    - K–1: "Make a ring of 8 counters on the table. Which hops land on every counter before you are

back?" (1, 3, 5 and 7.)

- 2–3: "Draw 12 dots like a clock. Which hops from 1 to 11 reach every dot without starting again?" (1, 5, 7 and 11; hop 7 is hop 5 the other way.)
- 4–5: "Find a ring and a hop that make 7 pieces, each one a five-pointed star." (35 dots with hop 14 or 21. Hops 7 and 28 give 7 pentagons.)

Opener: the round table (8 minutes)

A soft logic puzzle on the tables as children arrive, then acted out on the five chairs. The seats are counted clockwise, the way the hops are, and the answer's note passed 2 seats along is the launch's hop of 2, so the room goes straight from the puzzle into the launch. An adult reads the clues aloud at any table that wants it.

The puzzle (most of the room). Five animals sat around a round table. Write each animal's name in its seat: Bear, Fox, Owl, Rabbit, Turtle. Clockwise is the way the arrow goes; to go 2 seats clockwise, call the next seat *one* and the seat after it *two*.

1. Rabbit sat in the star seat.
2. Fox sat 2 seats clockwise from Owl.
3. Bear sat 2 seats clockwise from Fox.
4. Owl sat right next to Turtle.

Answer: clockwise from the star seat, Rabbit, Bear, Owl, Turtle, Fox. It is the only answer, and every clue is needed (checked by computer, round-table_check.py).

Hint ladder (give one step, then walk away):

1. "Which clue tells you exactly where one animal sat? Write that one first." (Rabbit, in the star seat.)
2. "Try Owl in the first seat after the star seat. Where do Fox and Bear go?" (Fox three seats after the star, and then Bear lands in the star seat, which is Rabbit's. Owl three seats after the star fails too: Fox lands in the star seat. So Owl is two or four seats after the star.)
3. "Owl is two or four seats after the star. In which one can Turtle sit next to Owl?" (Two. With Owl four seats after the star, Owl's neighbours are Bear and Rabbit.)

Easier (younger children, or anyone stuck at step 1). Four seats. Turtle sat in the star seat. Owl sat across the table from Turtle. Fox sat 1 seat clockwise from Turtle. *Answer:* clockwise from the star seat, Turtle, Fox, Owl, Bear. *Done?* "Turtle passes a note 2 seats clockwise, and on it goes. Who never gets it?" (Fox and Bear: K–1 Problem 1's hop 2 on 4 dots.)

Harder (older children, or anyone done in two minutes). Six seats and six animals, with Mouse. Mouse sat in the star seat. Fox sat 2 seats clockwise from Owl. Bear sat 2 seats clockwise from Fox. Owl sat right next to the star seat. Bear sat across the table from Rabbit. *Answer:* clockwise from the star seat, Mouse, Owl, Rabbit, Fox, Turtle, Bear. (Owl on the other side of the star seat puts Bear straight across from the star seat, and then Rabbit would need Mouse's seat.) *Done?* "Mouse passes a note 2 seats clockwise, and on it goes. Which animals never get it?" (Owl, Fox and Bear: they make their own loop. This is the midpoint's question, met early.)

Early finishers. "Cross out clue 4. Find every answer that fits now. Write a new clue that makes it exactly one answer again." (Two answers: the one above, and Rabbit, Fox, Turtle, Bear, Owl. "Bear sat right next to Rabbit" fixes it, checked, and so do others.)

On the floor (minute 6). Five children who solved it take the signs and sit where the answer says, Rabbit on the star seat. The room checks each clue out loud, thumbs up or down, counting the seats with the sitters. They stay seated: they are the launch's five dots. Bridge question: "Rabbit passes the note 2 seats clockwise, and each animal who gets it does the same. Does every animal get it before it is back with Rabbit?"

The hour

Time	Everyone	Jobs
0:00–0:08	**Opener: the round table.** Puzzle on the tables as children arrive; at minute 6, five solvers sit on the chairs and the room checks the clues.	Solvers check each other's sheets; adults give one hint step at a time.
0:08–0:11	**Launch at the chairs** (guide, "Launch: one hop"). The organizer gives Rabbit the note: "A hop of 2 moves it two seats clockwise." Rabbit passes it, everyone counting "one, two"; the organizer draws the line on the launch board. Once, on purpose, the organizer counts the seat the note is on as "one": "Was that 2 seats?" The note goes Owl, Fox, Bear, Turtle, back to Rabbit, and the board shows a five-pointed star. "Every animal got it. Does that always happen? Find out at your tables." Practise the reset signal: "Hop stop!", both hands on heads.	Sitters pass; everyone counts out loud; the organizer draws.
0:11–0:19	**Round 1.** K–1 Problems 1 and 2 (pages 1–2). 2–3 Problem 1 (page 1). 4–5 Problem 1 (page 1).	K–1: **Hopper** counts aloud; **Marker** marks each landing and says "back". 2–3: **Drawer**, and **Counter**, who counts each hop on the dots. 4–5: a pair and a **Referee**. Swap each ring.
0:19–0:27	**Round 2.** K–1 Problems 3 and 4 (pages 3–4). 2–3 Problem 3 (page 3), guess first; Problem 2 (page 2) if there is time. 4–5 Problem 2, first row (page 2), then Problem 3 (page 3), without drawing. At 0:25 each host asks the guide's sharing question, "Name a ring and a hop where you had to start again. Could you have told before you drew it?", and picks one child to bring a result to the chairs.	Same jobs, swapped. 4–5 rotates the referee.
0:27–0:32	**Back to the floor.** Add the sixth chair. Six new sitters, two from each table; the star seat stays. Pass the note by 2 from the star seat: who never gets it? (Three sitters.) "How many notes would we need so that everyone gets one?" (2.) "Which pass reaches everyone?" (1, or 5: try 5.) The 4–5 child says their rule so far; nobody says whether it is right. Then: "You may change tables now." Hosts turn their table cards round.	One child from each table says their result. A child who changes tables is welcomed by one who stayed, who reads the card aloud.
0:32–0:40	**Round 3.** K–1 Problem 4 if not done, then Problem 8 (page 8, the picture ring). 2–3 Problem 4 (page 4): cut and join wheels (pages 8–9), then decode; three codes is enough. A pair absorbed by the stars may do Problems 6–7 (pages 5–6) first. 4–5 Problem 7 (page 6), the rule in writing, about 3 minutes; then cut and join wheels (pages 10–11).	Wheels: one child cuts the outer wheel, the other the inner, and they join one wheel together. Then **Coder** turns and reads, **Writer** writes. A child still cutting at 0:40 gets a spare wheel and takes their own pages home.
0:40–0:48	**Round 4.** K–1 Problems 9 and 10 (pages 9–10), or Problems 6–7 (pages 6–7, the ring of 7) for a pair still keen on stars. 2–3 Problem 5 (page 4), codes for each other, then Problem 8 (page 7). 4–5 Problem 4 (page 4), crack the codes, then Problem 6 (page 5). At 0:46 each host asks: "Which hop or setting undoes yours?"	Same jobs, swapped. 4–5: the referee writes a secret code for the pair (the guide's crack race) once the codes are cracked.
0:48–0:55	**Floor finale: the yarn star** (step by step below). Everyone in one circle on the floor, 12 people if possible. Pass 3: guess how many starts, then roll; three squares. Pass 5: guess, then roll; one twelve-pointed star, lifted to waist height.	Jay and Rose, or two children from the 4–5 table, are floor captains: they point to the person the ball goes to.
0:55–1:00	"Hop stop!" Yarn into a bag; counters back in cups; pages collected; wheels and wheel pages go home.	One child per table collects pages; each pair counts one cup.

The yarn star, step by step. Everyone sits in one circle on the floor, adults joining or stepping out to make 12. The star sheet goes in front of the first holder, who keeps the loose end. *Pass 3:* the room guesses how many starts it will take. The holder rolls the ball to the person 3 along, everyone counting "one, two, three" from the

next person. Each catcher holds the yarn with one hand and rolls the ball on. When it is back with the first holder: "Back!" An adult cuts the yarn, and the holder keeps both ends. "Who has no yarn?" Start again there. Three starts make three squares. Everyone lays the yarn down and an adult clears it. *Pass 5:* guess, then roll. One start reaches everyone; everyone stands and lifts the twelve-pointed star to waist height, then "Hop stop!" and lay it down. If time is short, skip pass 3.

The finale for other numbers of people. Make the circle 12 if you can. Otherwise:

People in the circle	Pass that needs restarts	Pass for one big star
8	2 (2 starts: two squares)	3
9	3 (3 starts: three triangles)	2
10	4 (2 starts: two five-pointed stars)	3
11 or 13	none: every pass reaches everyone	3 or 5
14	4 (2 starts: two seven-pointed stars)	3
15	3 (3 starts: three pentagons)	4
16	4 (4 starts: four squares)	5

With 11 or 13, run one star, ask the 4–5 children why no pass needs a restart (11 and 13 do not split into equal piles), then have an adult join or step out for the restart pass. All rows checked by computer in round-table_check.py.

If it goes sideways

- A table finishes early: hand out its early-finisher card.
- A child counts the dot the counter is on as "one": "Put your finger on the next dot and say one" (guide, K–1 hint 1), nothing more.
- A pair is stuck on a drawing: "You are back. Point to a dot with no line. Start there in a new colour" (guide, Problem 4 hints).
- 2–3 writes "starts = hop" for every guess on 12 dots: leave it; hop 5 will surprise them, and that is the point (guide).
- 4–5 counts "pieces" as the parts that touch: pieces means starts (guide, Grades 4–5).
- Decoding comes out as nonsense: the usual mistake is going inner to outer, which codes it again. "Find V on the outer wheel. Which inner letter is beside it?" (guide, hint 2).
- Cutting runs long: a spare wheel at 0:40; their own pages go home. Reading the codes slows a pair down: the Writer says the letters and the host writes.
- K–1 runs out of attention: the guide's pass-the-counter game at the table (4 children and the host make 5; pass by 2; then the host steps out and pass by 2 again).
- The room gets loud: "Hop stop!", then go straight to the next floor activity.
- No floor space for the finale: stand round the biggest table and pass a counter instead of yarn; nothing is drawn, so the host draws the lines on the launch board as the counter moves.
- The yarn tangles or rolls away: an adult carries the ball to the right person, and the count goes on.
- Jay and Rose: the use log has no Week 4 entry. If they have met star drawing at home or in the app (Clockwork Gates), make them floor captains at 0:27 and 0:48, and checkers at the 2–3 table who may say "try again" but not why.

How it went

(Add a dated line after the session.)